

MeLearn

Machine learning that learns how to improve itself.

An autonomous agent that writes, evaluates, and improves recommendation models—without a human in the loop.

KUAI RAND - REQUIRED

+0.003944

PRIMARY SCORE GAIN



MeLearn cleared the required margin on validation—using the official evaluator.

EXPERIMENTS

30

of a 50-experiment cap

TOTAL COST

\$3.65

2.46M model tokens

HUMAN INTERVENTIONS

0

after the run started

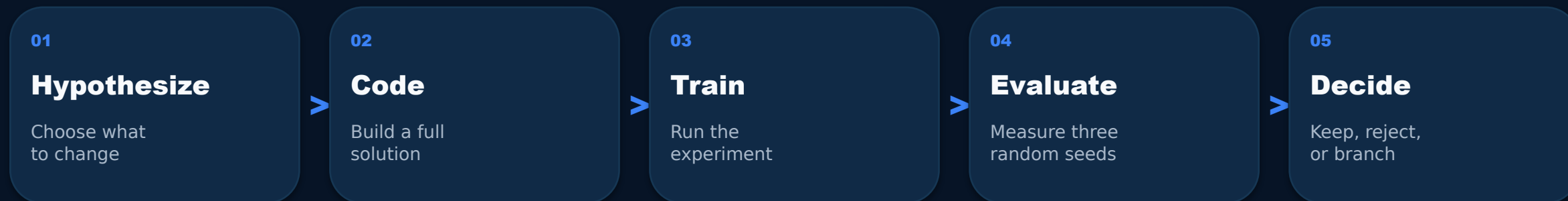
GPU-HOURS

0

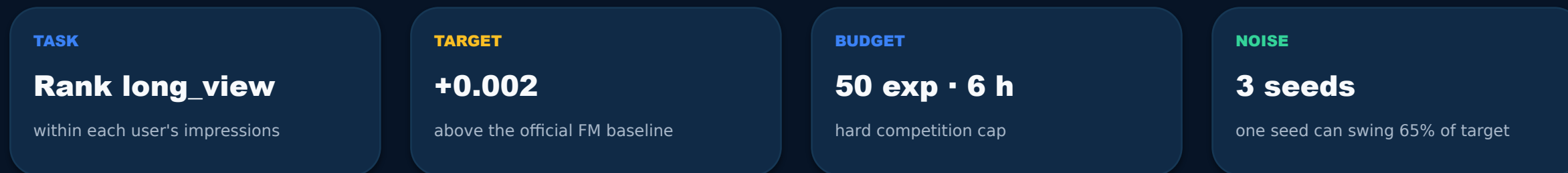
CPU-only record run

Improving a recommender is still a manual loop.

Tiny ranking gains demand disciplined experimentation—not one lucky training run.



The benchmark makes that loop unforgiving



MeLearn automates the whole research loop—not just model inference.

One command. Then leave it.

A multi-hour autonomous run, compressed here into one visible experiment cycle.

melearn · autonomous run

```
> python -m agent --dataset pure

[run] ledger loaded · 26 completed experiments

[uct] selected parent node 24

[idea] add a small DeepFM ensemble member

[write] solutions/027_deepfm_ensemble.py

[eval] seed 1/3  seed 2/3  seed 3/3

[score] primary 0.605493 · gain +0.003944

[ledger] verdict: KEPT · continue search
```

HOURS COMPRESSED FOR DEMO

WHAT THE AGENT DOES

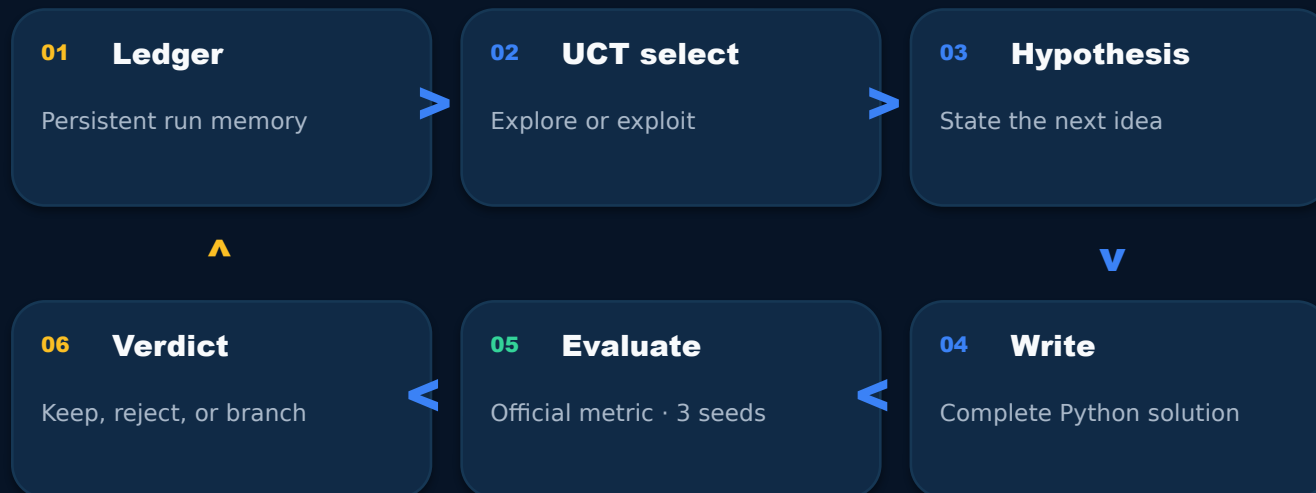
- 01 Read the research history
- 02 Choose a promising branch
- 03 Write a standalone solution
- 04 Evaluate across three seeds
- 05 **Persist the result and repeat**

WINNER · EXPERIMENT 27

The harness makes autonomy trustworthy.

The model proposes experiments; the surrounding system owns truth, limits, and memory.

THE CLOSED RESEARCH LOOP



The ledger is the memory. The model starts every turn clean.

UCT can return to an idea abandoned twenty iterations ago.

MECHANICAL GUARDRAILS

TEST

Labels are never exposed

SCORE

Official evaluator stays read-only

SEEDS

Every candidate runs three times

INPUT

Invalid or duplicate outputs are blocked

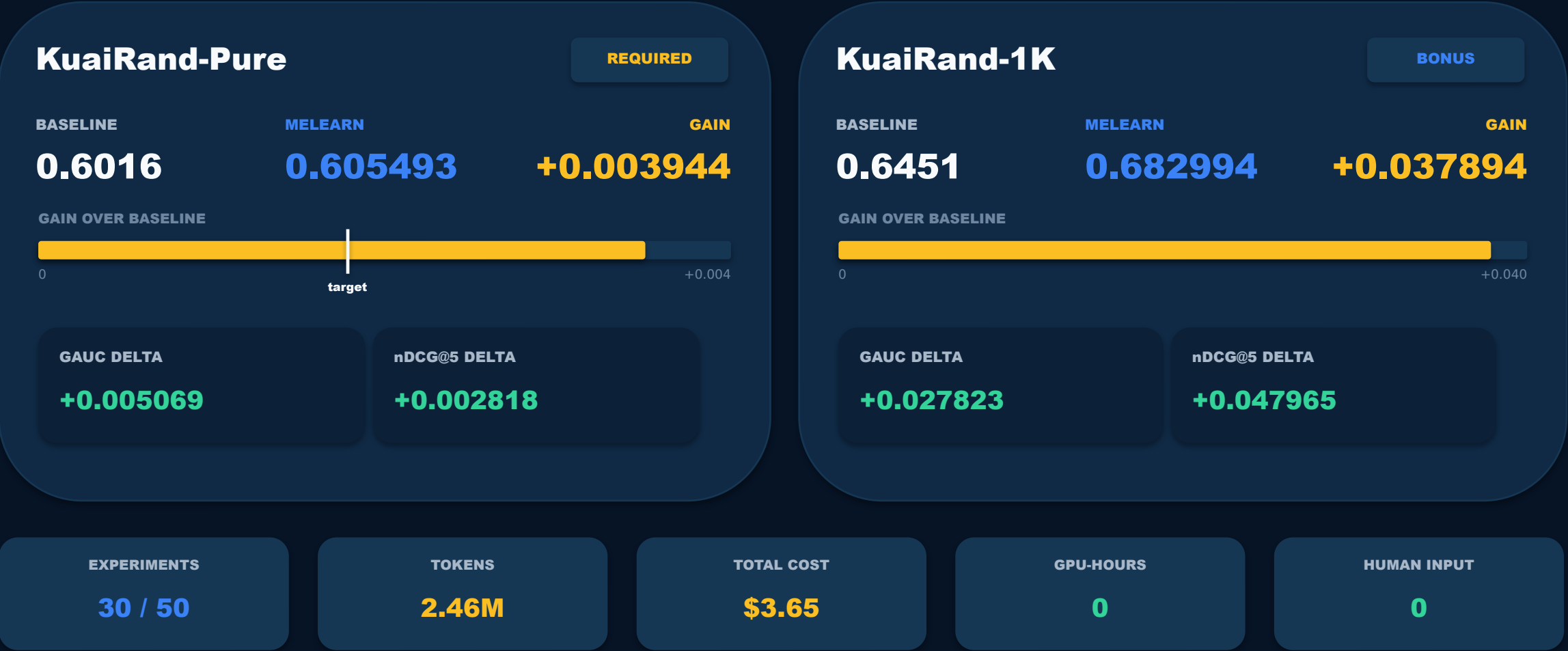
FAIL

Crashes return as actionable text

The agent cannot reason around these rules.

MeLearn cleared the required target.

Validation-best results · primary score = mean(GAUC, nDCG@5) · official evaluator unmodified



About 10× the overlap. About 10× the gain.

The same agent and harness exposed where the recommendation signal can—and cannot—transfer.



MeLearn did not just find a better model.
It found where the learnable signal lives.